# CMPSC 448 Midterm Project: LLM Identification
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**Problem definition**:

- The goal of this project was to determine whether a machine learning model can identify which large language model produced a given text response.

- Three models were used: ChatGPT, Gemini, and DeepSeek.

- Research questions RQ1 and RQ2 were investigated.

**Dataset curation**:

- There are 100 prompts used for collecting data on the three LLM
- The prompts are in six categories:
    - Factual QA
    - Coding
    - Math
    - Creative writing
    - Summarization
    - Opinions/advice

- Data cleaning through regex:
    - ChatGPT's image embedding that were included as a part of the response
    - Gemini's links that is included as a part of the response

**Implementations**:

**CNN architecture**

- The CNN reads sequences of up to 200 tokens. Each token is embedded into a 128-dimensional vector. Three Conv1d layers run in parallel with kernel sizes 3, 4, and 5, each producing 100 feature maps. Global max-pooling reduces each feature map to a single number, so the concatenated vector is 300 dimensions. Then dropout at 0.4 and a linear layer to 3 classes.

**RNN architecture**

- The RNN uses the same 128-dim embeddings and 200-token limit. It's a single bidirectional LSTM layer with hidden size 96. The final forward and backward hidden states are concatenated into a 192-dim vector. Then dropout at 0.4 and a linear layer to 3 classes.

**Training detail**:

- Both CNN and RNN share the same setup:
    - Framework
        - PyTorch
    - Optimizer
        - Adam
    - Learning rate
        - 1e-3
    - Weight decay
        - 1e-4
    - Loss
        - Cross-entropy
    - Gradient clipping
        - max norm 1.0
    - Batch size
        - 16
    - Max sequence len
        - 200
    - Epochs
        - 8 with early stopping
    - Early stop metric
        - Validation macro-F1
    - Dropout
        - 0.4
    - Seeds run
        - 1 through 5

**The result**:

- The chance for three classes is 1/3 = 0.333.

**How the results are presented**:

- RQ1:

| Model | Accuracy | Macro F1 |
|---|---|---|
| CNN | 0.756 | 0.756 |
| RNN | 0.511 | 0.444 |

  - Both CNN and RNN above chance. The CNN does noticeably better, which makes sense on short text with a small training set convolutional filters pick up on local character patterns (bold markers, header tags, code fence symbols) more reliably than a recurrent net that has to figure out which parts of the sequence matter.
  - The most common error for both models is DeepSeek being predicted as ChatGPT. They both use markdown headers and some bold labels, and neither use LaTeX nor tables. Gemini is rarely confused with the other two since it contains heavy LaTeX and table usage is very distinctive.

- RQ2:

| Model | Prompt only | Response only | Prompt + Response |
|---|---|---|---|
| CNN | 0.333 | 0.756 | 0.822 |
| RNN | 0.333 | 0.511 | 0.511 |

  - Prompt-only accuracy is exactly at chance for both models. That confirms the same prompts really were used across all three models and the prompt itself has no identifying signal since you can't predict which LLM will answer a question just from the question.
  - Adding the prompt to the response doesn't help meaningfully. The RNN data is unchanged. The CNN goes from 0.756 to 0.822, but since prompt-only is at random, that gap is most likely seed variance instead of a real effect.

**Any in-depth analyses/experiments**:

- Tested for length and vocabulary of the responses from the three different models and found that Gemini provides the most meaningful characters while DeepSeek provides the least, but this could be because DeepSeek tends to provide short and concise answers by default.

- Also Tested for how each model formats their response and it clearly shows that Gemini uses the most types of marker for formatting its answer. It also shows that DeepSeek relies heavily on headers and code blocks. At last ChatGPT shows the plainest responses which have almost no bold letters, no Latex format, and only a few tables.

**Lessons&Experience you learned in this project**:

- I learned that there are many distinct differences in how different LLM generate their responses as they use different formats of embedding and wording.

- I learned that formatting is important in both generating and distinguishing responses as I tested and debugged many formats of prompts for generating and codes to clean off the excessive information that aren't necessary inside the responses.

**What would I do if I had more time**:

- I would expand the amount of prompts per model to get better confidence intervals for results.

- I would test more models so that the project could identify more details and be able to distinguish more styles.

- I would want to look into RQ3 and RQ4 so I could improve the project more.